

TANISH MISRA

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EDUCATION

Purdue University

West Lafayette, IN

B.S. Computer Science (Machine Intelligence & Security Concentrations), Statistics Minor

Graduating December 2028

- Coursework: Problem Solving & OOP, Programming in C, The Data Mine, Linear Algebra, Discrete Mathematics

EXPERIENCE

Toyota Motor North America

Plano, TX

Software Engineering Co-op, Automation Team, Connected Experiences Division (CXD)

May 2026 – August 2026

- Built an AI pipeline converting 1,100+ in-vehicle infotainment test cases into automation-ready scripts, cutting authoring time from 3+ hours to 25 seconds per case using Pydantic v2 schema validation.
- Engineered a swappable LLM backend across Azure, Vertex, and Ollama for rapid model upgrades, with few-shot retrieval and an evaluation framework scoring schema validity, grounding, and reference fidelity.

The Data Mine, Purdue University

West Lafayette, IN

Undergraduate Data Science Researcher, 12 completed projects

January 2026 – Present

- Benchmarked pandas vs Polars on 500K+ row flight datasets, measuring read performance at scale and comparing explicit vs implicit API design patterns for filtering, selection, and type coercion.
- Created a color-coded US state map with GeoPandas and Folium, ingesting Census boundary data, reprojecting coordinate systems, normalizing state names for a spatial join, and rendering styled interactive maps.

Q Spark Group (QSG)

Remote

Data, Analytics, and AI Intern

March 2024 – July 2025

- Trained a Transformer NER model on 1M+ synthetic records across 3 PII classes, improving accuracy from 22% to 90% by retraining on misclassified cases.
- Applied the model to 250,000+ SQL records to classify sensitivity and assign access permissions company-wide.

Purdue Electric Racing

West Lafayette, IN

Firmware Developer, Software Team

August 2025 – Present

- Implemented git-hash validation firmware in embedded C for 7 STM32 node types, verifying version consistency across 90 CAN messages at boot.
- Classified firmware mismatches into 3 severity levels to eliminate manual repo inspection during fault diagnosis.

Electric Vehicle Club

West Lafayette, IN

Software Developer, Autonomous Team

August 2025 – Present

- Co-developed an OpenCV vision node within a 9-node ROS2 autonomous kart system, detecting lane edges via color segmentation and tracking edges across frames.
- Computed steering angles using triangle geometry on detected edge coordinates, reducing full-frame scans through stateful edge tracking.

PROJECTS

Telogify (F1 Telemetry Intelligence) | *LangGraph, PostgreSQL, Anthropic API, React*

August 2026

- Built an F1 telemetry platform generating 10 data-grounded insights per race weekend through a LangGraph ReAct agent bound to 12+ deterministic Postgres query tools.
- Eliminated LLM hallucination by filtering output against a 60-phrase blocklist and validating every cited number against query results before persisting.

Candle for Choroideremia (CHM) | *FastAPI, PostgreSQL, pgvector, OpenAI API*

April 2026

- Designed a research dashboard ingesting 29 clinical trials and 475 publications, enabling plain-English patient queries through vector similarity search with medical safety guardrails.
- Adopted by CureCHM after user acceptance testing by 3 blind patients and featured in a press release.

Securify | *LangGraph, RoBERTa, OpenAI API | AlgoFest 2026 Hackathon Semi-Finalist*

April 2026

- Architected a 4-node LangGraph security pipeline redacting PII through a fine-tuned RoBERTa NER model with 0.940 micro F1 across 10 entity classes.
- Blocked prompt injections via 18 regex filters and validated PII-free output across medical, financial, and HR data.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, TypeScript

AI/ML & Data: LangChain, LangGraph, HuggingFace, Anthropic API, OpenAI API, PyTorch, OpenCV, PostgreSQL

Developer Tools: Git/GitHub, Docker, Claude Code CLI, Codex, REST APIs, React, Vercel, Railway